

Introduction

The Structural Kinetic Energy Conversion System (SKECS) is an innovative technology designed to take advantage of the mechanical oscillations of skyscraper counterweights to generate clean electricity. By integrating a rack and pinion system with these existing stabilizers, the technology converts the kinetic energy from wind-induced building movement into a passive power source. This system offers a high **theoretical efficiency of 50-70%** (Gieras, 2017), significantly outperforming traditional urban renewable options like solar panels and wind turbines. Beyond its efficiency, the SKECS provides a sustainable solution with a low carbon footprint, because it doesn't require any extra land to use and produces no emissions. While installation costs could be estimated to be in the low millions (Tang, 2011), the system presents a viable economic alternative for dense urban environments where space is limited. This research shows how repurposing pre-existing structural infrastructure can reduce a building's reliance on the power grid and make it much more efficient.



Figure 2: Taipei 101 Tuned Mass Damper and Exterior of Building (Video Thumbnail)

Purpose

The motivation behind the SKECS system comes from the immediate need for clean energy as global warming increases the number of extreme storms. While rising wind speeds offer a significant energy source, traditional renewable energy sources such as wind farms or solar panels commonly require large amounts of land that are unavailable in dense cities, driven by the need to capture the wasted kinetic energy from existing building counterweight movements, providing a sustainable, passive power source makes the building more energy efficient, cleanly.

Generating Electricity From The Passive Movement of a Tuned Mass Damper to Improve Skyscraper Energy Efficiency

Phase 1 ~ Model building construction
The base is assembled from two 2x4s and a wood platform, as well as the floors of the tower, which are all made from wood platforms. Supported by 4 aluminum rods.



Figure 1: Initial tower base

Phase 2 ~ Motor & ratcheting system mounting
Motors are assembled then screwed into the top platform of the tower. Ratcheting system designed in AutoCAD then mounted near motors to allow secondary counterweights to drop when they reach the top.



Figure 5: Original Prototype Design

Phase 3 ~ Counterweight mounting and wiring
Both main and secondary counterweights are mounted on tower, the main on a cross axle. Wires are then run up the tower's support beams to motors.



Figure 3: Motors mounted on top wooden platform



Figure 2: Unfinalized tower

Phase 4 ~ Testing & Data Analysis
Testing completed by wiring motors into a voltage sensor to detect voltage generated when secondary counterweights are released from the ratcheting system and spin the motors.

Testing



Figure 6: New 600rpm motors freshly wired/soldered on large custom printed support piece.



Figure 8: Testing Process Utilizing Multimeter and Motor Setup

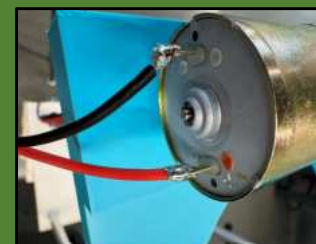


Figure 8: New solder on 600rpm motors - For testing, multimeter attached to the motor prongs.

Testing conducted by letting each side counterweight fall once each then simultaneously, spinning the motors, to determine the average voltage produced by each cycle of dropping. Multimeter used at the prongs of each motor to avoid wire length resistance issues, which were later fixed.

Results

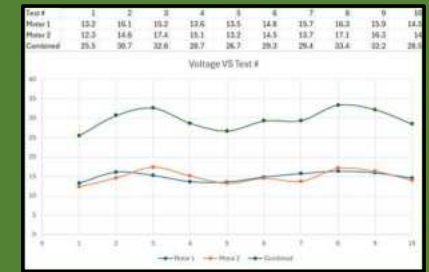


Figure 8: Data table of average voltage values collected from each motor.

Data for two motors and the average voltage generated between them was gathered through testing. The data collection yielded voltage values of around 30V, found by averaging all of the values.

Conclusion

The research was successful to a certain extent, qualitatively, but not quantitatively. This is because the mechanical processes were found to be too complicated using this method of energy generation to store enough potential energy to turn the motors. Because of this, the quotative data obtained does not account for the difficulty in raising the counterweights in the first place. This kind of system *may* be viable at a large scale if different, more compact and efficient means of energy conversion were to be used, such as a **triboelectric generator**. The largest improvement to be made is to use more **prefabricated components** to make pursuing the main goal of the project easier.

References

- Zin-qi, Z. (n.d.). A review on energy-regenerative suspension systems ... A Review on Energy-Regenerative Suspension Systems for Vehicles. https://www.laeng.org/publication/WCE2013/WCE2013_pp1889-1892.pdf
- U.S. Energy Information Administration. (2022, December 21). Geothermal explained. Geothermal power plants. <https://www.eia.gov/energyexplained/geothermal/geothermal-power.php>
- Gieras, J., Fellow, L., Kucharski, A., & Plechowski, J. (2017). Performance characteristics of a shake flashlight Performance Characteristics of a Shake Flashlight. <https://doi.org/10.1109/IEEMDC.2017.8002088>
- Tang, X., & Zuo, L. (2011). Simulation and Experiment Validation of Simultaneous Vibration Control and Energy Harvesting from Buildings using Tuned Mass Dampers. In 2011 American Control Conference, 2011 American Control Conference (p. 3134). AAC. <https://doi.org/10.1109/ACC.2011.5990784>